# Database Concepts, Importing Data & Mastering the SELECT Statement

Learn how data is organized in enterprise databases and retrieve it like a Data Engineer.

# Today's Agenda

✓ Quick Recap
✓ Database vs Schema vs Table
✓ Understanding Dot Notation  (Database.Schema.Table)
✓ Creating an HR Schema
✓ Creating Tables using Schema
✓ Importing Data from CSV Files
✓ SELECT *
✓ Specific Columns
✓ Aliases (AS)
✓ CONCAT()
✓ DISTINCT
✓ TOP
✓ Assignment
✓ SQL Best Practices

# Quick Recap

✓ Learned What a Table Is
✓ Created Our First Table
✓ Inserted Sample Data
✓ Verified the Data
✓ Retrieved Required Information Using SELECT

# What is a Database?

A database is a organized collection of related data that allows information to be stored, managed, and retrieved efficiently.

**Imagine You Own a Company...**

As your company grows, you'll store information such as:

- 🧑‍💼 Employees
- 🛒 Customers
- 📦 Products
- 💰 Orders
- 📊 Sales
- 📋 Projects



🗄️ Database
DataEngineerLabs

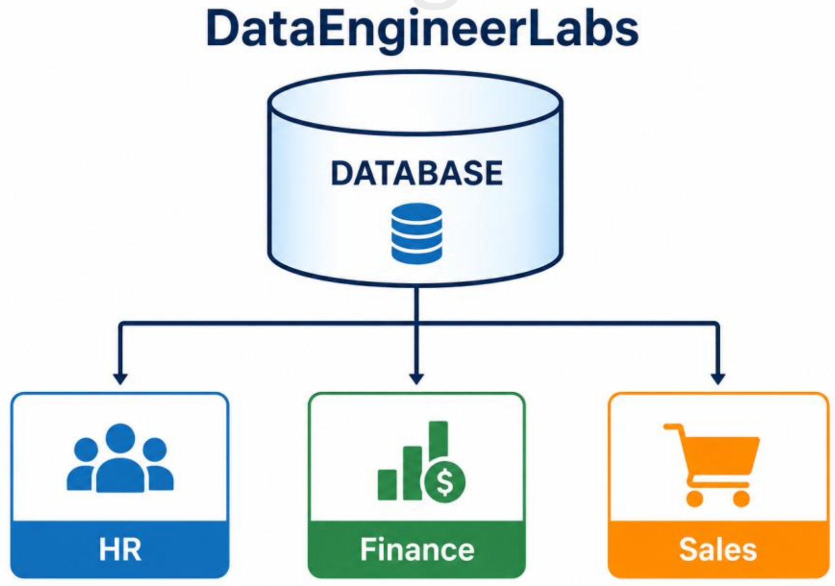
- 🧑 Employees
- 🛒 Customers
- 📦 Products
- 📋 Orders
- 🚀 Projects

# What is a Schema?

A Schema is a logical grouping of related database objects within a database. It helps organize, secure, and manage data more effectively.

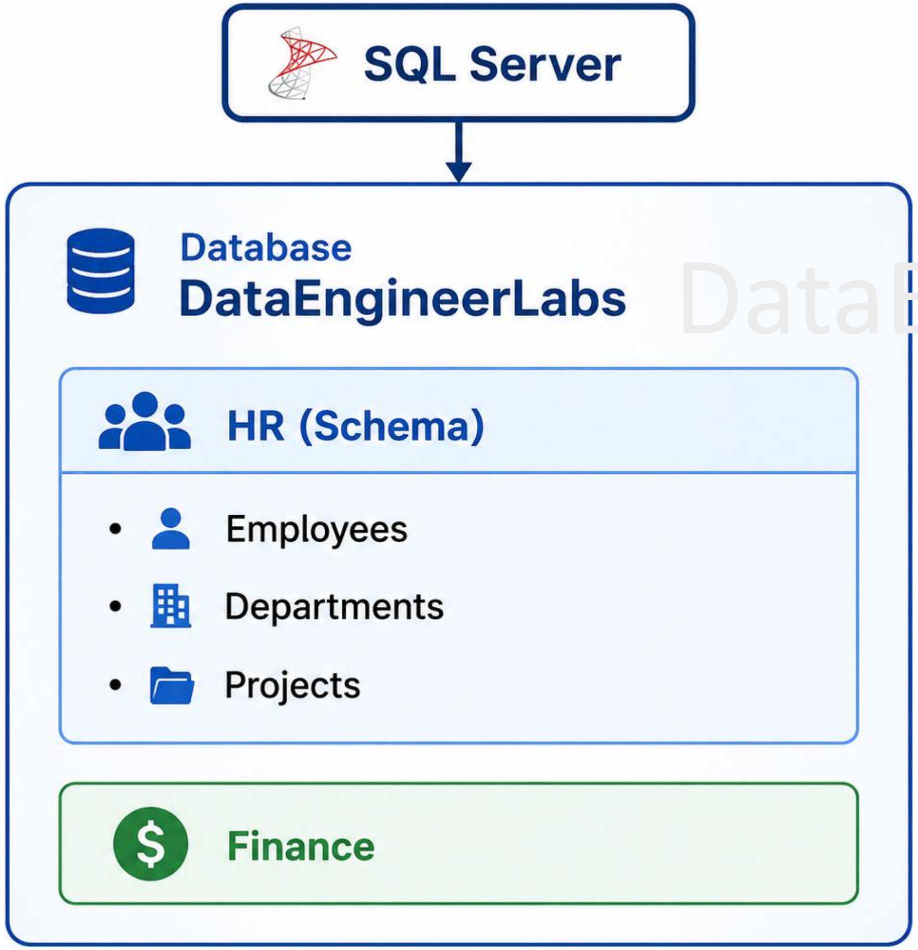
**Example:** HR, Finance and Sales are common schemas in enterprise databases.





# What is a Table?

A table is an organized collection of related information stored in rows and columns.





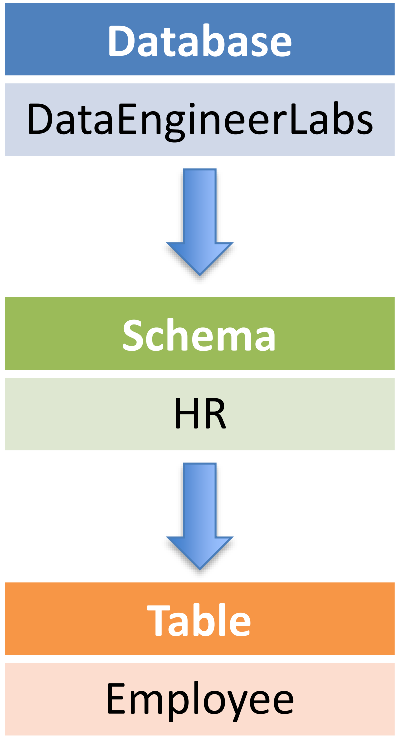

## Employees Table

| ID | Name | Department | Salary |
|---|---|---|---|
| 1 | John Smith | HR | 60000 |
| 2 | Emily Johnson | IT | 75000 |
| 3 | Michael Brown | Finance | 70000 |
| 4 | Sarah Davis | Marketing | 65000 |
| 5 | David Wilson | Sales | 68000 |

# Understanding Dot Notation

In SQL Server, every database object has an address, just like every house has a postal address.
Instead of using a street, city, and country, SQL Server identifies objects using:
**Database.Schema.Table**
Each part of the name tells SQL Server exactly where to find the table.

```
SELECT * FROM DataEngineerLabs.HR.Employees;
```



# Create HR Schema and Tables

```
USE DataEngineerLabs;

GO

CREATE SCHEMA HR;

GO
```

# Create table and load data from CSV

Execute the table creation script

Create_Employees.sql

Import Employees.csv using BULK INSERT



```
BULK INSERT HR.Employees
FROM ‘path\to\Employees.csv’
WITH (
        FIRSTROW = 2,
        FIELDTERMINATOR = ',’,
         ROWTERMINATOR = '\n’,
        TABLOCK
 );
```

# Mastering the SELECT Statement

1. Let's say HR wants to see every piece of information stored about employees
2. HR only needs the employee's First Name, Last Name, and Salary
3. The HR report should display more user-friendly column names
4. The HR Manager wants a Full Name column instead of separate First Name and Last Name columns
5. Management wants to know how many departments have/had employees in the company
6. The HR team wants to preview only the first few employee records.

# Episode Summary

✓ Database, Schema, Tables
✓ Dot Notation
✓ Created First Schema
✓ Created table and loaded data into tables
✓ Retrieved all columns using SELECT *
✓ Selected specific columns
✓ Renamed columns using AS
✓ Combined values using CONCAT()
✓ Removed duplicates using DISTINCT
✓ Limited results using TOP

# Assignment

Complete the following SQL tasks using the **HR** schema.

**Question 1**
Display **all records** from the HR.Employees, HR.Projects and HR.Departments table.

**Question 2**
Display only the following columns:
EmployeeID
FirstName
LastName
DepartmentID



**Question 3**
Create a report showing:
Employee ID
Full Name (using CONCAT())
Email Address
Rename the Full Name column as **Employee Name**.

**Question 4**
Display a list of **unique Department IDs** available in the Employees table.

**Question 5**
Create following report using HR.Employee table :

| Employee ID | Employee Name | Job Title | Salary |
|---|---|---|---|

# Data Engineer Tip

💡 Data Engineer Best Practice

✓ Avoid using SELECT * in production.
✓ Retrieve only the columns you need.
✓ Use meaningful aliases.
✓ Format SQL for better readability.
✓ Practice every query yourself.

# Thank You

Keep Building. Keep Experimenting